
LoRA ANALYSIS VIA STABLE RANK

Nikita Okhotnikov
okhotnikov.nv@phystech.edu

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1 Introduction

Recent advances in large language models (LLMs) and multimodal models (MLLM) induce a growing number of various downstream tasks attempted to be solved with a single pretrained model. Despite that the performance of such models vastly improved over last years, there are still tasks that cannot be handled in zero-shot setting and some others just benefit a lot from additional fine-tuning. However, the size of LLMs with cutting-edge performance keeps growing and even best open-sourced models nowadays contain hundreds of billions of parameters[2][12][13][4]. Thus, full finetuning of such models is infeasible in most cases and parameter efficient finetuning (PEFT) methods are required. The most widespread PEFT option is low-rank adaptation (LoRA)[6]. Since the release of the original paper, the approach has been actively studied (CITE SOMETHING HERE), however it still lacks comprehensive theoretical background of expressivity, generalization and expected behaviour. ...

2 Related work

The most general estimate of expressive power of LoRA is presented in [14]. The paper provides rigorous constructive proof of the lower bound for the rank of adapter to accurately represent the target model, for a single linear layer, multi-layer perceptron and even slightly modified transformer models. Being a notable limitation this result is unfortunately as practical as the first universal approximation theorem [5] and its follow-ups [1][3][8]. While the existence of a matrix of specific rank that satisfy the required conditions might shed some light on the theoretical boundaries, it provides no guarantees that such matrix could be achieved with standard gradient optimization techniques. Moreover, the authors of [9] showed that the matrix *stable rank* [10] of a LoRA matrices is much smaller (actually close to 1) than the regular rank in downstream tasks training regime that drastically limits the actually reachable subspace of matrices of a given rank and makes the previously discussed theoretical boundaries even less applicable.

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[7] presents the upper and lower bounds for the generalization gap of asymmetric low-rank adapters [15] in terms of rank and sample complexity. Generalization gap is defined as the difference between empirical risk and true risk of the model on a given data distribution. The upper bound in the paper is proportional to the $\frac{\sqrt{r}}{N}$, where r is the adapter rank and N is the number of training samples. That is based on the worst case estimates of r -rank matrices and suggests that the potential generalisation might worsen with higher adapter rank. However, it could be potentially tightened if we constrain the stable rank of the resulting LoRA perturbation ΔW .

[THERE'S NEW NOT FULLY CHECKED RESULT WITH $\sim \sqrt{sr(\Delta W)} \implies$ NO GROWTH OF THE GENERALIZATION GAP WITH THE INCREASE OF THE RANK AS LONG AS THE STABLE RANK ASSUMPTION HOLDS]

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[SOME WORDS ABOUT RANK COLLAPSE (STABLE RANK) IN ATTENTION LAYERS [11] AS AN ADDITIONAL MOTIVATION THAT REGULAR RANK IS NOT THE RIGHT TOOL TO FORM THE THEORY AROUND]

3 Main part (THINK OF THE TITLE)

3.1 Single matrix approximation

Lemma 1. Let the A be a target matrix and $\hat{A}$ - an estimate. Let $S_A = \mathbf{sr}(A)$, $S_{\hat{A}} = \mathbf{sr}(\hat{A})$, $S_{\hat{A}} \leq S_A$ and $\sigma_1(\hat{A}) = \alpha \sigma_1(A)$ $\alpha \geq 0$, where $\sigma_1(\cdot)$ is the first (largest) singular value of corresponding matrix. Then:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right)$$

Proof.

$$\begin{aligned} \mathbf{sr}(A) &= \frac{\|A\|_F^2}{\sigma_1(A)^2} \implies \|A\|_F = \sigma_1(A) \sqrt{\mathbf{sr}(A)} \\ \|A - \hat{A}\|_F &\geq \left| \|A\|_F - \|\hat{A}\|_F \right| \quad \text{triangle inequality} \\ \|A - \hat{A}\|_F &\geq |\sigma_1(A) \sqrt{S_A} - \sigma_1(\hat{A}) \sqrt{S_{\hat{A}}}| = \sigma_1(A) |\sqrt{S_A} - \alpha \sqrt{S_{\hat{A}}}| \\ \mathbf{1. } \alpha &\leq 1: \quad \|A - \hat{A}\|_F \geq \sigma_1(A) (\sqrt{S_A} - \sqrt{S_{\hat{A}}}) \\ \mathbf{2. } \alpha &\geq 1: \quad \|A - \hat{A}\|_F \geq \|A - \hat{A}\|_{op} \geq |\sigma_1(A) - \sigma_1(\hat{A})| = \sigma_1(A)(\alpha - 1) \\ \|A - \hat{A}\|_F &\geq \sigma_1(A) \min_{\alpha \geq 1} \max (\sqrt{S_A} - \alpha \sqrt{S_{\hat{A}}}, \alpha - 1) \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right) \end{aligned}$$

□

Lemma 2. (Constructive lower bound) Assuming the conditions of the previous lemma, let A_R be the optimal unconstrained rank- R approximation of A with singular values $\sigma_1, \dots, \sigma_R$. Let $\overline{S_A} = \mathbf{sr}(A_R)$ Then there exists $\hat{A}$ such that $\mathbf{rank}(\hat{A}) \leq R$, $\mathbf{sr}(\hat{A}) = S_{\hat{A}} < R$, $S_A < R$ and

$$\|A - \hat{A}\|_F^2 = \sum_{i=R+1}^D \sigma_i^2 + (1 - \gamma)^2 \sum_{i=2}^R \sigma_i^2$$

Where $\gamma \in [0, 1]$ is defined as:

$$\gamma = \begin{cases} 1 & \text{if } \overline{S_A} \leq S \\ \sqrt{\frac{S-1}{\overline{S_A}-1}} & \text{if } \overline{S_A} > S \end{cases}$$

Proof.

1. Let the SVD of A be $\sum_{i=1}^D \sigma_i u_i v_i^T$. The minimal error for any rank- R matrix is bounded by the tail $\sum_{i=R+1}^D \sigma_i^2$. We focus on constructing $\hat{A}$ within the subspace of the top R singular vectors to minimize the additional error. Let $\hat{A} = \sum_{i=1}^R \hat{\sigma}_i u_i v_i^T$.

2. We aim to minimize $\sum_{i=1}^R (\sigma_i - \hat{\sigma}_i)^2$ subject to $\frac{\sum_{i=1}^R \hat{\sigma}_i^2}{\hat{\sigma}_1^2} \leq S$. If $\overline{S_A} \leq S$, we set $\hat{\sigma}_i = \sigma_i$ (yielding $\gamma = 1$) and the penalty is 0. If $\overline{S_A} > S$, we must reduce the stable rank. The most efficient way to do this (minimizing Euclidean distance) is to preserve the largest component $\hat{\sigma}_1 = \sigma_1$ (maximizing the denominator) and uniformly shrink the tail components $\hat{\sigma}_i = \gamma \sigma_i$ for $i = 2, \dots, R$.

3. The stable rank condition becomes equality:

$$\frac{\sigma_1^2 + \sum_{i=2}^R (\gamma \sigma_i)^2}{\sigma_1^2} = S, \quad 1 + \gamma^2 \frac{\sum_{i=2}^R \sigma_i^2}{\sigma_1^2} = S$$

Note that $\overline{S_A} = 1 + \frac{\sum_{i=2}^R \sigma_i^2}{\sigma_1^2}$. Thus:

$$1 + \gamma^2 (\overline{S_A} - 1) = S \implies \gamma^2 = \frac{S - 1}{\overline{S_A} - 1}$$

4. The squared error on the top R components is:

$$(\sigma_1 - \sigma_1)^2 + \sum_{i=2}^R (\sigma_i - \gamma \sigma_i)^2 = (1 - \gamma)^2 \sum_{i=2}^R \sigma_i^2$$

Adding the tail error $\sum_{i=R+1}^D \sigma_i^2$ gives the result. $\square$

3.2 Fully connected network approximation

3.2.1 Notation

Let $FNN_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L)$ be fully connected neural network with L layers of width D and ReLU activation functions. $W_l \in \mathbb{R}^{D \times D}$ – weight matrices $b_l \in \mathbb{R}^D$ – bias vectors.

In this work we use

$$\text{target FNN } \bar{f} := FNN_{\bar{L},D}(\cdot; (\bar{W}_l)_{l=1}^{\bar{L}}, (\bar{b}_l)_{l=1}^{\bar{L}}),$$

$$\text{frozen FNN } f_0 := FNN_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L),$$

$$\text{adapted FNN } f := FNN_{L,D}(\cdot; (W_l + \Delta W_l)_{l=1}^L, (\hat{b}_l)_{l=1}^L)$$

where $b_l, \bar{b}_l, \hat{b}_l \in \mathbb{R}^D$, $W_l, \bar{W}_l, \Delta W_l \in \mathbb{R}^{D \times D}$, but $\Delta W_l = A_l B_l$, where $A_l \in \mathbb{R}^{D \times R}$ and $B_l \in \mathbb{R}^{R \times D}$ and $R \leq D - \bar{L}$. Additionally, we assume and $\bar{L} = L$.

3.2.2 Expressive power of FNN

Theorem 1. Let $x \in \mathbb{R}^d$ be the random input with covariance matrix $\Sigma = \mathbb{E}[xx^T]$. Let $E_l = \bar{W}_l - W_l$ be the weight discrepancy at layer l . Let $\sigma_{l,i}$ be the singular values of E_l . Assume $\hat{b}_l = \bar{b}_l$. Then there exist adapters ΔW_l of rank $\leq R$ and $\forall l \in \bar{1}, \bar{L}$ $\text{sr}(\Delta W_l) \leq S$ such that the expected approximation error is bounded by:

$$\mathbb{E} \|f(\mathbf{x}) - \bar{f}(\mathbf{x})\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\bar{W}_k\|_2 \right) \mathcal{T}_l$$

Where β is the magnitude of the parameters and the input

$$\beta = \max_{i \in [L]} \left(\underbrace{\left(\prod_{j=1}^i \|\bar{W}_j\| \right)}_{\text{Input Term}} \sqrt{\text{tr}(\Sigma)} + \underbrace{\sum_{j=1}^i \left(\prod_{k=j+1}^i \|\bar{W}_k\| \right) \|\bar{b}_j\|}_{\text{Bias Term}} \right)$$

and $\mathcal{T}_l$ is the error from **Lemma 2**:

$$\mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + (1 - \gamma_l)^2 \sum_{j=2}^R \sigma_{l,j}^2}, \text{ with } \gamma_l = \min \left(1, \sqrt{\frac{S-1}{\text{sr}(E_l) - 1}} \right)$$

Proof.

1. Linearization: Let h_l be the output of layer l in the adapted model, and $\bar{h}_l$ in the target model.

$$h_l = \text{ReLU}((W_l + \Delta W_l)h_{l-1} + \bar{b}_l)$$

$$\bar{h}_l = \text{ReLU}(\bar{W}_l \bar{h}_{l-1} + \bar{b}_l)$$

We define the error vector

$$e_l = h_l - \bar{h}_l = \text{ReLU}((W_l + \Delta W_l)h_{l-1} + \bar{b}_l) - \text{ReLU}(\bar{W}_l \bar{h}_{l-1} + \bar{b}_l)$$

As **ReLU** is 1-Lipschitz

$$\|e_l\|_2 \leq \|((W_l + \Delta W_l)h_{l-1} + \bar{b}_l) - (\bar{W}_l \bar{h}_{l-1} + \bar{b}_l)\|_2$$

Add and subtract $\overline{W}_l h_{l-1}$:

$$\|e_l\|_2 \leq \|(W_l + \Delta W_l - \overline{W}_l)h_{l-1} + \overline{W}_l(h_{l-1} - \overline{h}_{l-1})\|_2$$

Let $M_l = W_l + \Delta W_l - \overline{W}_l$ be the approximation residual matrix for layer l .

$$\|e_l\|_2 \leq \|M_l h_{l-1} + \overline{W}_l e_{l-1}\|_2$$

2. Unrolling the recursion: By induction from $l = 1$ to L (with $e_0 = 0$):

$$\|e_L\|_2 \leq \sum_{l=1}^L \left\| \prod_{k=l+1}^L \overline{W}_k \right\|_2 \|M_l\|_2 \|h_{l-1}\|_2$$

3. Constructing the Adapter ΔW_l : We need to minimize $\|M_l\|_F = \|(W_l + \Delta W_l) - \overline{W}_l\|_F$. This is equivalent to approximating $E_l = \overline{W}_l - W_l$ with ΔW_l . Thus we can apply and this yields the Frobenius error:

$$\|M_l\|_F = \mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + (1 - \gamma_l)^2 \sum_{j=2}^R \sigma_{l,j}^2}$$

Since $\|M_l\|_2 \leq \|M_l\|_F$, we use $\mathcal{T}_l$ as the bound for the matrix norm.

4. Handling the intermediate norms: We look at $\mathbb{E}\|e_L\|_2$. The term depends on $\mathbb{E}\|h_{l-1}\|_2$. $\|h_{l-1}\|_2 \leq (\prod_{j=1}^{l-1} \|\overline{W}_j\|_2) \|\mathbf{x}\|_2$. For the expectation, $\mathbb{E}\|\mathbf{x}\|_2 \leq \sqrt{\mathbb{E}\|\mathbf{x}\|^2} = \sqrt{\text{tr}(\Sigma)}$.

We want to bound $\|\overline{h}_i\|_2$.

$$\begin{aligned} \|\overline{h}_1\| &\leq \|\overline{W}_1\| \|\mathbf{x}\| + \|\overline{b}_1\| \\ \|\overline{h}_2\| &\leq \|\overline{W}_2\| \|\overline{h}_1\| + \|\overline{b}_2\| \end{aligned}$$

Substitute the bound for Layer 1:

$$\begin{aligned} \|\overline{h}_2\| &\leq \|\overline{W}_2\| (\|\overline{W}_1\| \|\mathbf{x}\| + \|\overline{b}_1\|) + \|\overline{b}_2\| \\ \|\overline{h}_2\| &\leq \underbrace{\|\overline{W}_2\| \|\overline{W}_1\| \|\mathbf{x}\|}_{\text{Input Contribution}} + \underbrace{\|\overline{W}_2\| \|\overline{b}_1\| + \|\overline{b}_2\|}_{\text{Accumulated Bias Contribution}} \end{aligned}$$

By induction, the magnitude of the activation at layer i is bounded by:

$$\|\overline{h}_i\| \leq \left(\prod_{j=1}^i \|\overline{W}_j\| \right) \|\mathbf{x}\| + \sum_{j=1}^i \left(\prod_{k=j+1}^i \|\overline{W}_k\| \right) \|\overline{b}_j\|$$

Taking the expectation and the maximum over layers we get the β defined in theorem conditions 5. Gather up the terms:

$$\mathbb{E}\|f(\mathbf{x}) - \overline{f}(\mathbf{x})\|_2 = \mathbb{E}\|e_L\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\overline{W}_k\|_2 \right) \mathcal{T}_l$$

□

3.2.3 Expressive power of transformer

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3.3 Upper-bound for LoRA generalization gap

[HERE'S CURRENTLY UNSOLVED PROBLEM WITH THE PROOF!]

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In this section we use specifically the setting of [7] paper. Consider a depth- T linear network with 1-Lipschitz activation functions $\sigma(\cdot)$, pre-trained weights θ_{pre} with $W_{pre}^{(t)}$ weight matrices for each layer. LoRA is applied at some layers via matrices $\Delta W^{(t)} = B^{(t)} A^{(t)}$, where $B^{(t)}$ is random and fixed, $A^{(t)}$ is trainable. Let $\mathcal{F}$ be the resulting random class of functions. Loss is bounded in $[0, 1]$

Theorem 2. Assuming that for each adapted layer t , with probability at least $1 - \epsilon$

$$\|\Delta W^{(t)}\|_{op} \leq \Lambda, \quad \mathbf{sr}(\Delta W^{(t)}) \leq S, \quad \mathbf{sr}(M) := \frac{\|M\|_F^2}{\|M\|_{op}^2}$$

$\forall \delta \in (0, 1]$, the following holds with probability at least $1 - \delta$:

$$G = \sup_{f \in \mathcal{F}} |R(f) - R_N(f)| \leq 4 \min \left\{ 1, \frac{\sqrt{(cT+1)(\sum_t (m_t + n_t))S}}{\sqrt{6N}} \right\} + \sqrt{\frac{8 \log \frac{2}{1-\sqrt{1-\delta}}}{N}}$$

where $m_t \times n_t$ is the shape of $\Delta W^{(t)}$, $R(f)$ – true risk, $R^N(f)$ – empirical risk

Proof. The proof follows similar 3-stage structure as the original rank-based theorem.

Step 1 - Lipschitz constant bound

Consider the mapping

$$LoRA : \Omega \times \mathcal{G} \rightarrow C([0, 1]^d, \mathbb{R}^D),$$

$$\mathcal{G} := \{(\Delta W^{(1)}, \dots, \Delta W^{(T)}) : \|\Delta W^{(t)}\|_{op} \leq \Lambda, \mathbf{sr}(\Delta W^{(t)}) \leq S\} \subset \mathbb{R}^q$$

where $\mathcal{G}$ – the space of vectors of LoRA parameters that satisfy the conditions of the theorem,

$f_\theta(\cdot) \in C([0, 1]^d, \mathbb{R}^D)$ – the model function with adapter with weights $\theta \in \mathcal{G}$ applied

Let constant c be so that $\forall t : \|W_{pre}^{(t)}\|_{op} \leq c$.

Then $\|W^{(t)}\|_{op} \leq \|W_{pre}^{(t)}\|_{op} + \|\Delta W^{(t)}\|_{op} \leq c + \Lambda$

We define intermediate activations $h_t(x)$ recursively: $h_0(x) = x$, $h_t(x) = \sigma(W^{(t)}(h_{t-1}(x)))$, $t \in T$

Let us fix the parameter vectors θ and θ' , then as $\sigma(\cdot)$ is 1-Lipschitz, the following holds:

$$\|h_t^\theta - h_t^{\theta'}\| = \|\sigma(W^{(t)}h_{t-1}^\theta) - \sigma(W'^{(t)}h_{t-1}^{\theta'})\| \leq \|W^{(t)}h_{t-1}^\theta - W'^{(t)}h_{t-1}^{\theta'}\| \leq \|W^{(t)}\|_{op} \|h_{t-1}^\theta - h_{t-1}^{\theta'}\| + \|(W^{(t)} - W'^{(t)})h_{t-1}^{\theta'}\|$$

$$\|(W^{(t)} - W'^{(t)})h_{t-1}^{\theta'}\| \leq \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op} \|h_{t-1}^{\theta'}\|$$

$$\|h_t(x)\| \leq \|W^{(t)}\|_{op} \|h_{t-1}(x)\| \leq (c + \Lambda)^t \|x\| \leq (c + \Lambda)^t$$

$$\|h_t^\theta - h_t^{\theta'}\| \leq (c + \Lambda) \|h_{t-1}^\theta - h_{t-1}^{\theta'}\| + (c + \Lambda)^{t-1} \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op}$$

$$\|h_t^\theta - h_t^{\theta'}\| \leq \sum_{t=1}^T (c + \Lambda)^{T-t} (c + \Lambda)^{t-1} \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op} \leq T(c + \Lambda)^{T-1} \max_t \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op}$$

$$\|f_\theta(x) - f_{\theta'}(x)\| \leq T(c + \Lambda)^{T-1} \max_t \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op}$$

Thus $LoRA$ mapping defined above is Lipschitz with constant $L = T(c + \Lambda)^{T-1}$ if we consider $\mathcal{G}$ as a space with the metric $d(\theta, \theta') = \max_t \|\Delta W^{(t)} - \Delta W'^{(t)}\|_{op}$.

Step 2 - Covering number bound

Definition (Covering number):

Let $(V, \|\cdot\|)$ be a normed space.

Let $B(\rho, v_0) = \{v \in V : \|v - v_0\| \leq \rho\}$ be the ball with radius ρ and center $v_0 \in V$. $B(\rho) = B(\rho, 0)$

For any $K \in B$ and number $\varepsilon > 0$ an ε -covering of K is a finite set $\{v_1, \dots, v_N\} \subset K$ such that

$$K \subset \bigcup_{i=1}^N B(\varepsilon, v_i)$$

Then ε -covering number $\mathcal{N}(\varepsilon, K)$ of a compact set $K \subset V$ is defined as

$$\mathcal{N}(\varepsilon, K) = \min\{N : \exists \text{ an } \varepsilon\text{-covering of } K \text{ with size } N\} < \infty$$

$$\|M\|_{op} \leq \Lambda, \text{sr}(M) \leq S \implies \|M\|_F \leq \Lambda\sqrt{S}$$

$$\text{Let } \mathcal{M} = \{M : \in \mathbb{R}^{m \times n} : \|M\|_F \leq \Lambda\sqrt{S}\}$$

[PROBLEM FROM HERE, NEED TO FIX THE PROOF]

□

References

- [1] Yoshua Bengio and Olivier Delalleau. On the expressive power of deep architectures. In *International conference on algorithmic learning theory*, pages 18–36. Springer, 2011.
- [2] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jiawei Wang, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhua Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shutong Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Shengfeng Ye, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wanjia Zhao, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanhong Xu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025.
- [3] Ronen Eldan and Ohad Shamir. The power of depth for feedforward neural networks. In *Conference on learning theory*, pages 907–940. PMLR, 2016.
- [4] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lomakin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla,

Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shao-liang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collet, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget, Virginie Do, Vish Voleti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyan Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymier, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru,

- Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wencheng Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024.
- [5] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- [6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021.
- [7] Anastasis Kratsios, Tin Sum Cheng, Aurelien Lucchi, and Hartz Sáez de Ocariz Borde. Sharp generalization bounds for foundation models with asymmetric randomized low-rank adapters, 2025.
- [8] Shiyu Liang and Rayadurgam Srikant. Why deep neural networks for function approximation? *arXiv preprint arXiv:1610.04161*, 2016.
- [9] Kai Lion, Liang Zhang, Bingcong Li, and Niao He. Polar: Polar-decomposed low-rank adapter representation. *arXiv preprint arXiv:2506.03133*, 2025.
- [10] Mark Rudelson and Roman Vershynin. Sampling from large matrices: An approach through geometric functional analysis. *Journal of the ACM (JACM)*, 54(4):21–es, 2007.
- [11] Thiziri Nait Saada, Alireza Naderi, and Jared Tanner. Mind the gap: a spectral analysis of rank collapse and signal propagation in attention layers, 2025.
- [12] Kimi Team, Yifan Bai, Yiping Bao, Guanduo Chen, Jiahao Chen, Ningxin Chen, Ruijue Chen, Yanru Chen, Yuankun Chen, Yutian Chen, Zhuofu Chen, Jialei Cui, Hao Ding, Mengnan Dong, Angang Du, Chenzhuang Du, Dikang Du, Yulun Du, Yu Fan, Yichen Feng, Kelin Fu, Bofei Gao, Hongcheng Gao, Peizhong Gao, Tong Gao, Xinran Gu, Longyu Guan, Haiqing Guo, Jianhang Guo, Hao Hu, Xiaoru Hao, Tianhong He, Weiran He, Wenyang He, Chao Hong, Yangyang Hu, Zhenxing Hu, Weixiao Huang, Zhiqi Huang, Zihao Huang, Tao Jiang, Zhejun Jiang, Xinyi Jin, Yongsheng Kang, Guokun Lai, Cheng Li, Fang Li, Haoyang Li, Ming Li, Wentao Li, Yanhao Li, Yiwei Li, Zhaowei Li, Zheming Li, Hongzhan Lin, Xiaohan Lin, Zongyu Lin, Chengyin Liu, Chenyu Liu, Hongzhang Liu, Jingyuan Liu, Junqi Liu, Liang Liu, Shaowei Liu, T. Y. Liu, Tianwei Liu, Weizhou Liu, Yangyang Liu, Yibo Liu, Yiping Liu, Yue Liu, Zhengying Liu, Enzhe Lu, Lijun Lu, Shengling Ma, Xinyu Ma, Yingwei Ma, Shaoguang Mao, Jie Mei, Xin Men, Yibo Miao, Siyuan Pan, Yebo Peng, Ruoyu Qin, Bowen Qu, Zeyu Shang, Lidong Shi, Shengyuan Shi, Feifan Song, Jianlin Su, Zhengyuan Su, Xinjie Sun, Flood Sung, Heyi Tang, Jiawen Tao, Qifeng Teng, Chensi Wang, Dinglu Wang, Feng Wang, Haiming Wang, Jianzhou Wang, Jiaxing Wang, Jinhong Wang, Shengjie Wang, Shuyi Wang, Yao Wang, Yejie Wang, Yiqin Wang, Yuxin Wang, Yuzhi Wang, Zhaoji Wang, Zhengtao Wang, Zhexu Wang, Chu Wei, Qianqian Wei, Wenhao Wu, Xingzhe Wu, Yuxin Wu, Chenjun Xiao, Xiaotong Xie, Weimin Xiong, Boyu Xu, Jing Xu, Jinjing Xu, L. H. Xu, Lin Xu, Suting Xu, Weixin Xu, Xinran Xu, Yangchuan Xu, Ziyao Xu, Junjie Yan, Yuzi Yan, Xiaofei Yang, Ying Yang, Zhen Yang, Zhilin Yang, Zonghan Yang, Haotian Yao, Xingcheng Yao, Wenjie Ye, Zhuorui Ye, Bohong Yin, Longhui Yu, Enming Yuan, Hongbang Yuan, Mengjie Yuan, Haobing Zhan, Dehao Zhang, Hao Zhang, Wanlu Zhang, Xiaobin Zhang, Yangkun Zhang, Yizhi Zhang, Yongting Zhang, Yu Zhang, Yutao Zhang, Yutong Zhang, Zheng Zhang, Haotian Zhao, Yikai Zhao, Huabin Zheng, Shaojie Zheng, Jianren Zhou, Xinyu Zhou, Zaida Zhou, Zhen Zhu, Weiyu Zhuang, and Xinxing Zu. Kimi k2: Open agentic intelligence, 2025.
- [13] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025.
- [14] Yuchen Zeng and Kangwook Lee. The expressive power of low-rank adaptation, 2024.
- [15] Jiacheng Zhu, Kristjan Greenewald, Kimia Nadjahi, Hartz Sáez De Ocariz Borde, Rickard Br  el Gabrielsson, Leshem Choshen, Marzyeh Ghassemi, Mikhail Yurochkin, and Justin Solomon. Asymmetry in low-rank adapters of foundation models. *arXiv preprint arXiv:2402.16842*, 2024.